

The Program in Action

When you run *bitmapmessage.py*, the output will look like this:

```
Bitmap Message, by Al Sweigart al@inventwithpython.com
Enter the message to display with the bitmap.
> Hello!

Hello!Hello!Hello!Hello!Hello!Hello!Hello!Hello!Hello!Hello!Hello!He
  lo!Hello!Hello  l !He lo e      llo!Hello!Hello!Hello!Hello!He
    llo!Hello!Hello!Hello He lo H l !Hello!Hello!Hello!Hello!Hello H
el      lo!Hello!Hello!He      lo!Hello!Hello!Hello!Hello!Hel
      o!Hello!Hello      lo e lo!H ll !Hello!Hello!H l
        !Hello!He      llo!Hel  Hello!Hello!Hell ! e
          Hello!He      ello!Hello!Hello!Hello!Hell H
1      H llo! ell      ello!Hello!Hell !Hello el o
        lo!H l      ello!Hello!Hell ell !He o
          !Hello      llo!Hello!Hel el He o
            !Hello!H      lo!Hello!Hell l !H llo
              ello!Hel      Hello!He      H llo Hell
                ello!Hell      ello!H l      Hell !H l o!
                  ello!Hell      ello!H l o      o!H l H
                    lo!Hel      ello! el      o!Hel H
                      lo!He      llo! e      llo!Hell
                        llo!H      llo!      llo!Hello
                          llo!      ll      lo!Hell e
                            llo      l      l e
                              ll      l      H
Hello!Hello!Hello!Hello!Hello!Hello!Hello!Hello!Hello!Hello!Hello!He
```

How It Works

Instead of individually typing each character of the world map pattern, you can copy and paste the whole thing from <https://inventwithpython.com/bitmapworld.txt>. A line of 68 periods at the top and bottom of the pattern acts as a ruler to help you align it correctly. However, the program will still work if you make typos in the pattern.

The `bitmap.splitlines()` method call on line 43 returns a list of strings, each of which is a line in the multiline bitmap string. Using a multiline string makes the bitmap easier to edit into whatever pattern you like. The program fills in any non-space character in the pattern, which is why asterisks, periods, or any other character do the same thing.

The `message[i % len(message)]` code on line 51 causes the repetition of the text in `message`. As `i` increases from 0 to a number larger than `len(message)`, the expression `i % len(message)` evaluates to 0 again. This causes `message[i % len(message)]` to repeat the characters in `message` as `i` increases.

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1. `"""Bitmap Message, by Al Sweigart al@inventwithpython.com`
 2. `Displays a text message according to the provided bitmap image.`
 3. `View this code at https://nostarch.com/big-book-small-python-projects`